

Global CO₂ Emissions: Who Emits, How Much, and What's Changing

Fossil fuels & industry, 1750–2024

DATA FROM THE GLOBAL CARBON BUDGET (2025) VIA OUR WORLD IN DATA.

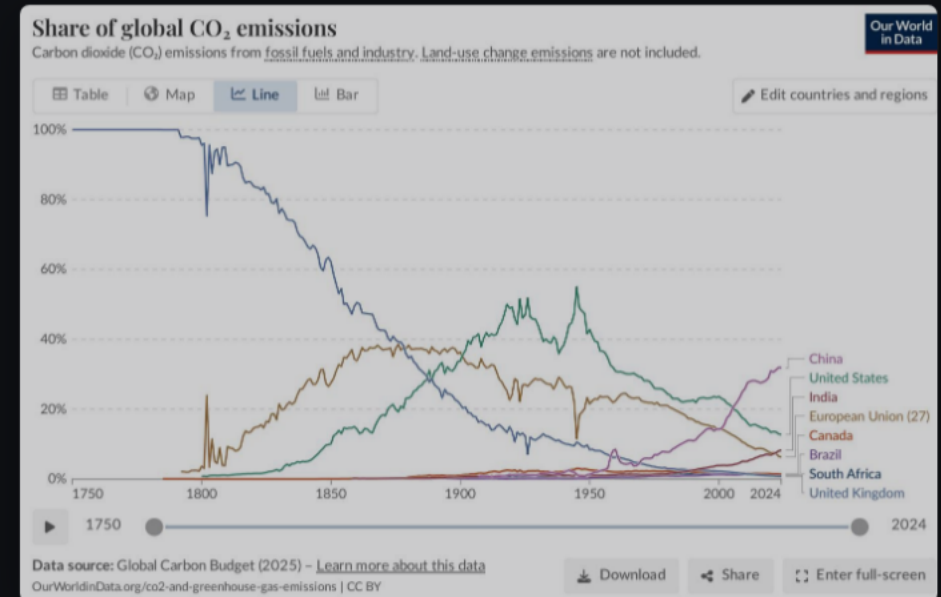
FOR POLICY ANALYSTS, CLIMATE NEGOTIATORS, AND SUSTAINABILITY RESEARCHERS.

Global CO₂ Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today

01

Global CO₂ emissions from fossil fuels and industry have grown dramatically since the Industrial Revolution.

- **Pre-Industrial:** Emissions were negligible.
- **1950 Acceleration:** Reached roughly 6 billion tonnes (Bt) per year.
- **1990 Milestone:** Nearly quadrupled to over 20 Bt.
- **Today:** Exceeds 35 Bt annually. Growth has slowed, but emissions have not yet peaked.



Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today

02

The distribution of global emissions has shifted profoundly over the last century.

- **1900:** Europe and the US accounted for >90% of global CO₂ emissions.
- **1950:** Still represented over 85% of the total.
- **Today:** Europe and the US account for less than 33%.
- **The Rise of Asia:** Led by China, Asia now dominates and produces roughly half of all global emissions.



Stacked Area Chart

Showing the historical rise of Asia's share and the relative decline of Europe and North America from 1900 to 2024.

China Now Emits More Than One-Quarter of Global CO₂

Comparing country-level shares of global CO₂ emissions in 2024 reveals massive concentration among a few top emitters.

~31%

China

Rising, growth slowing

~13%

United States

Declining

~8%

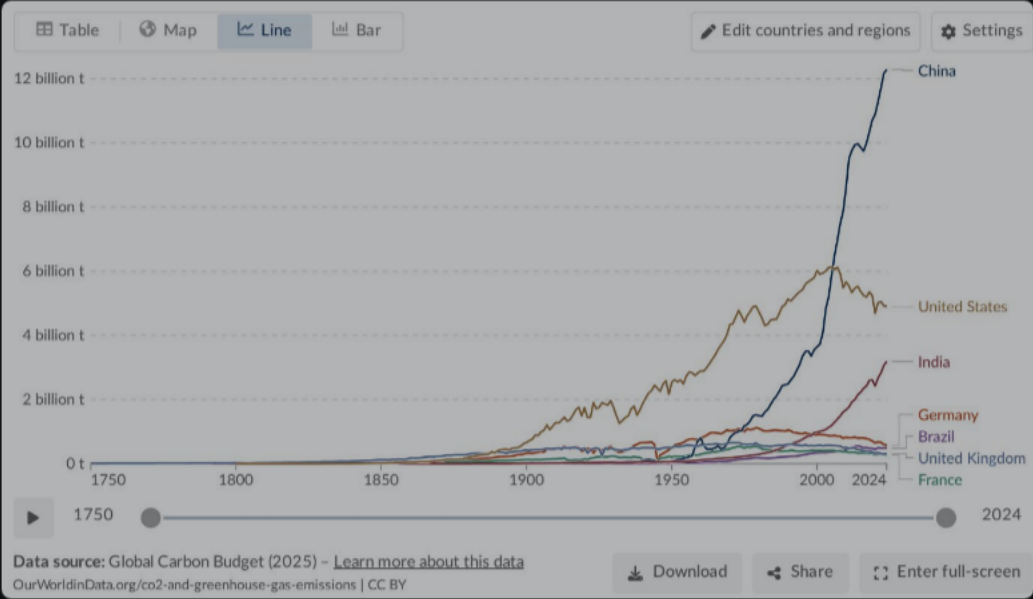
India

Rising

~6%

European Union (27)

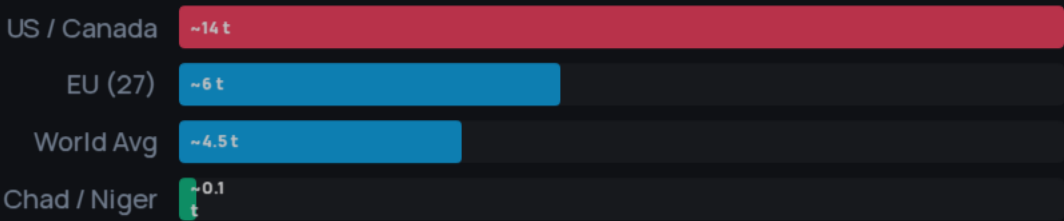
Declining



Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada

Per capita CO₂ emissions reveal stark global inequalities. The gap is enormous: the average American or Australian emits in under two days what the average person in Mali or Niger emits in an entire year.

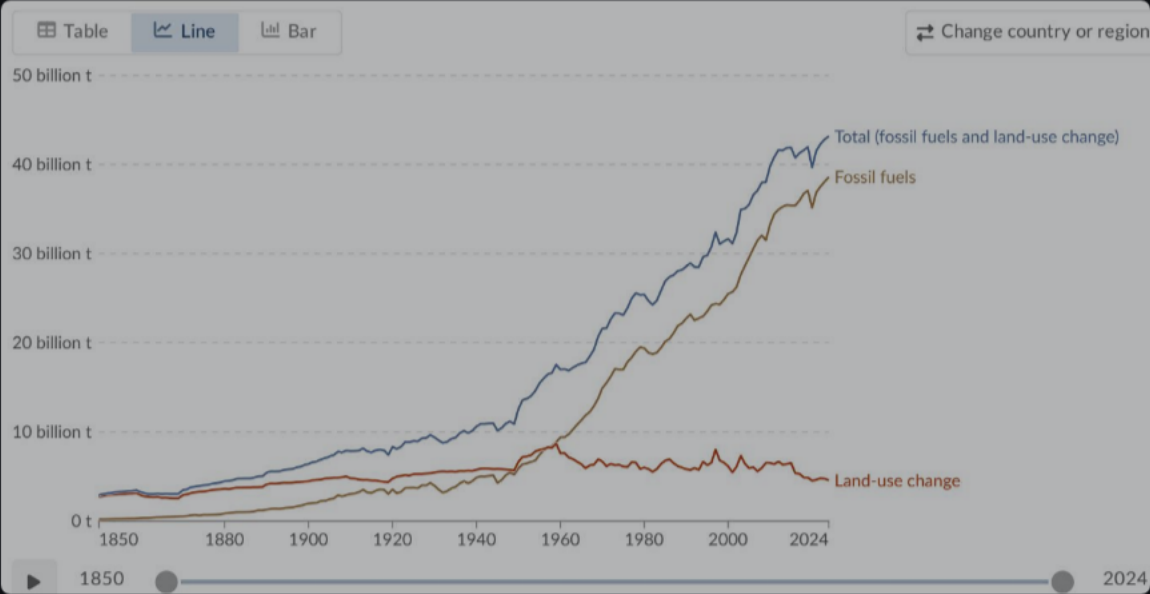
- **Highest Emitters:** Major oil-producing nations with small populations (e.g., Qatar, UAE).
- **Wealthy Populous Nations:** The US, Australia, and Canada have per capita emissions roughly 3x the global average.
- **150x Difference:** Between the lowest (Sub-Saharan Africa) and highest large-country emitters.



Comparative Horizontal Bar Chart (Data: ~2024)

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India

While China is the largest *total* emitter, its *per capita* emissions remain significantly below the United States.

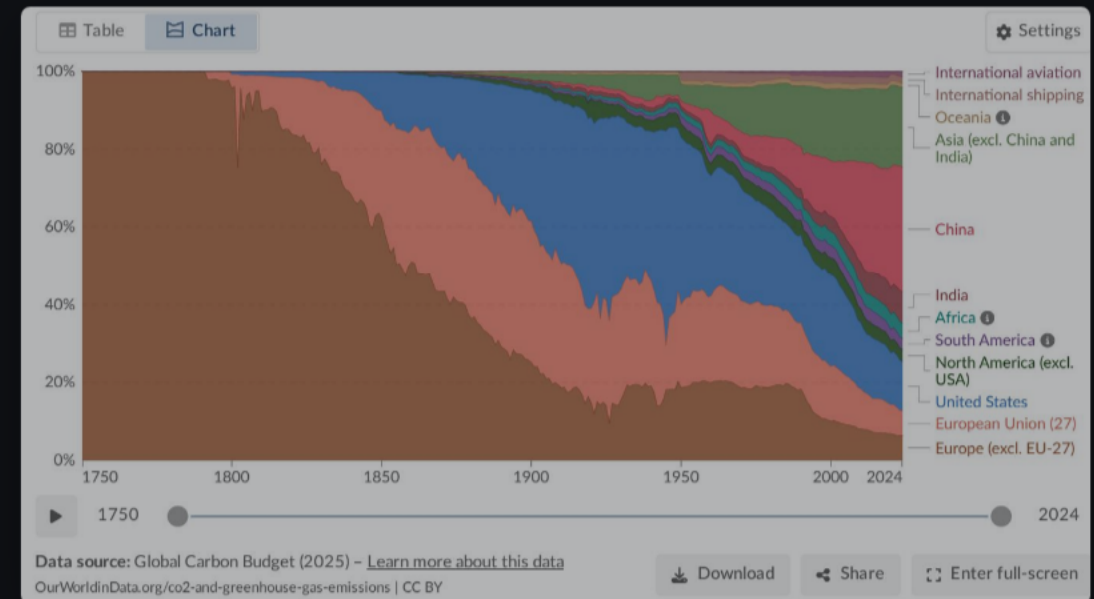


2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew

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The map of annual percentage change in CO₂ emissions for 2024 shows a highly mixed global picture.

- **Declines:** Many Western nations—including the US, much of the EU, and the UK—saw emissions drop.
- **Increases:** Russia, parts of Southeast Asia, and several African nations experienced growth.
- **Notable Spikes:** Some countries in Central Asia (e.g., Turkmenistan, Uzbekistan) saw large increases exceeding 10%.



Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US

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Even among wealthy nations with similar prosperity and living standards, per capita emissions can vary significantly based on their power grids.

- **Low Emitters (per capita):** France, the UK, and Portugal have successfully minimized per capita emissions (UK: ~4 t).
- **The Nuclear & Renewables Factor:** A much higher share of their electricity is generated from nuclear and renewable sources rather than fossil fuels.
- **Higher Emitters (per capita):** Germany and the US (US: ~14 t) maintain higher per capita emission rates due to heavier reliance on fossil-intensive energy infrastructure.



Diverging Energy Mix Bar Chart

Comparing the share of Nuclear/Renewables vs. Fossil Fuels in the electricity grids of the US, Germany, UK, and France alongside their per capita emissions.

Key Takeaways: Responsibility Is Shared but Unequal

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- **No peak yet:** Global emissions exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth.
- **Dramatic shift in responsibility:** Europe and the US dominated historically (>90% in 1900) but now account for less than one-third.
- **China's complex role:** China is the largest annual emitter (~31%), but its per capita emissions (~9 t) remain well below the US (~14 t).
- **Massive per capita inequality:** A 150x gap persists between the lowest emitters (Sub-Saharan Africa, ~0.1 t) and the highest large-country emitters.
- **Energy policy matters:** Countries with similar prosperity levels can have very different per capita emissions depending on their energy mix and technology choices.